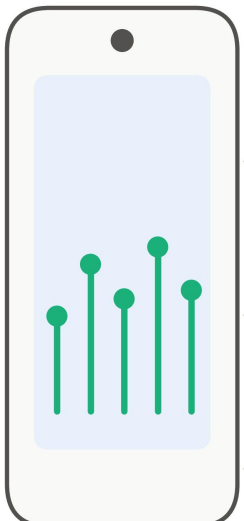
Smartphone AI phenotyping tool



- Accuracy
- Cost per plot
- **Breeding-cycle time**

Breeding simulation
5,760 tool profiles

Genetic gain
calibrated to 0.5%/yr

Adoption and NPV
25-year horizon

Cycle time leads; accuracy counts at low h^2

Equivalent years of cycle compression

Accuracy, 10% error cut (evidence)
6.9 months

Accuracy, 50% error cut (optimistic)
18.0 months

One year shorter breeding cycle
12 months

Worth measuring first in a pilot (EVPPI)

cycle time $7.0M vs accuracy $1.3M

Simulated East African common-bean program; plot-level h^2 = 0.10